SUNWAY UNIVERSITY

Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

Topic 03:
Cloud Computing

OUR VISION: TO BE A WORLD CLASS UNIVERSITY

NET3204
Distributed Systems

Chee Sun LIEW cheesunl@sunway.edu.my

Learning Outcome

- At the end of this lecture, you will be able to:
  - Explain the concept of cloud computing
  - List and describe the characteristics of cloud

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IBM


DIGITAL EQUIPMENT CORPORATION
PDP-8





source: www.apple.com



source: www.samsung.com

and now…




2014 CAN'T BE THE END OF THE PC! I'VE ONLY JUST LEARNT HOW TO USE THE DAMN THING!!!
© D.Fletcher for CloudTweaks.com

in the context as “a service”

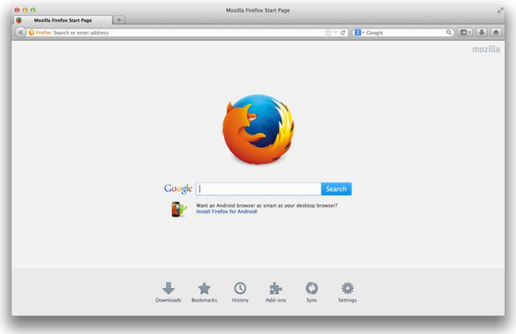
- What if ?
  - we pay someone to use their program
  - we pay someone to use their storage (hard disk)
  - we pay someone to run and maintain our IT infrastructure
  - we pay someone only
    when we’re using their services

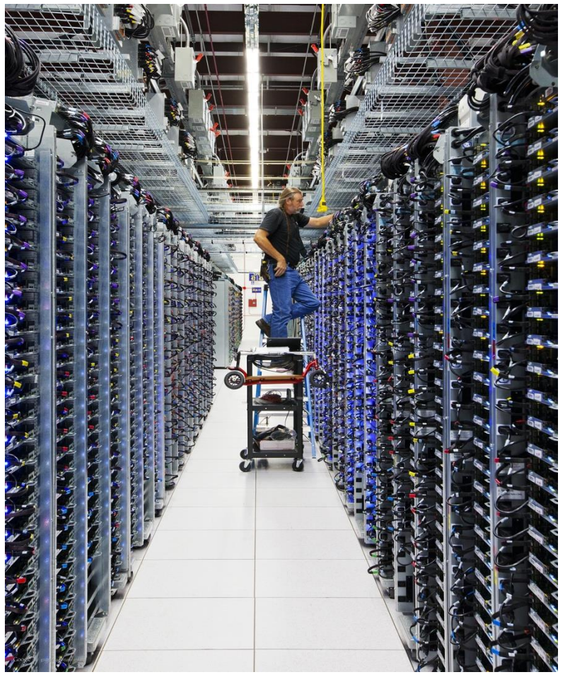
"… computing may someday be organised as a public utility just as the telephone system … The computer utility could become the basis of a new and important industry."

John McCarthy,
Computer Scientist, 1961





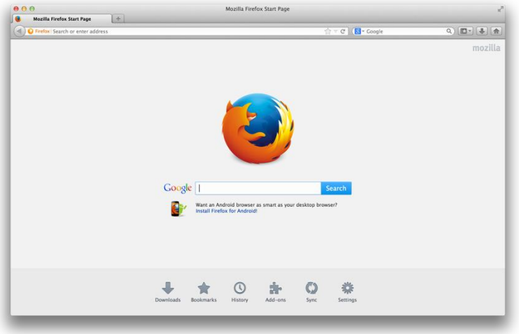

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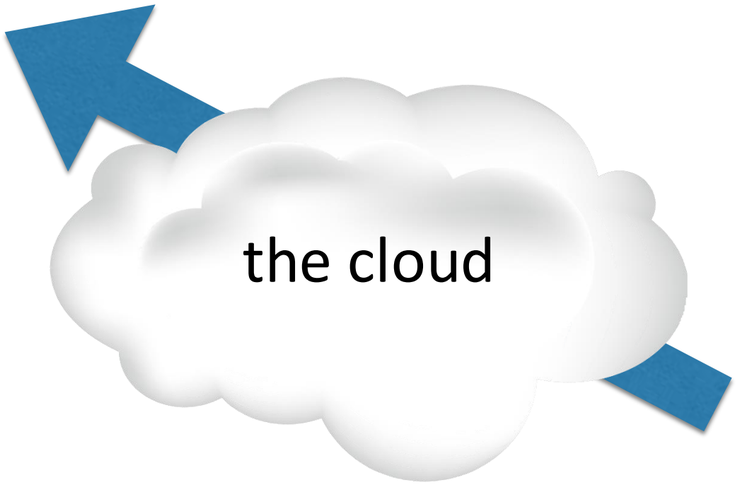


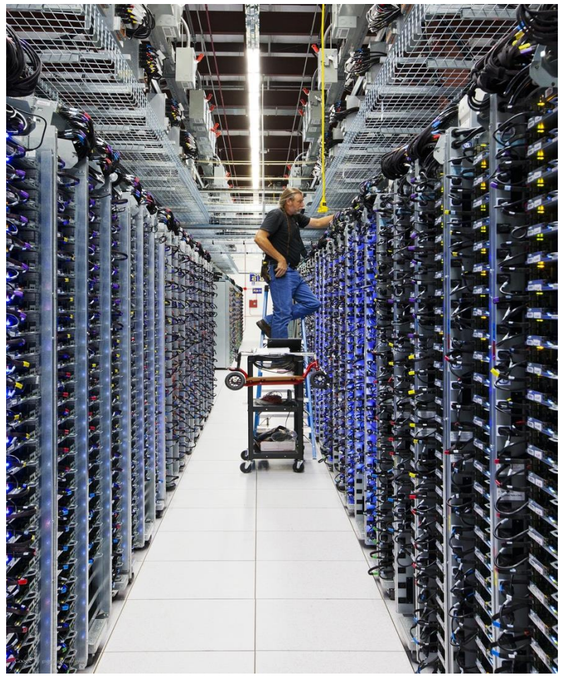
The idea of utility computing

- Computing resources as a metered services
  - the program you use (gmail)
  - the storage (dropbox)
  - the computing cycle (AWS EC2)

- that are
  - uniform
  - on-demand
  - available at all time
  - on a “pay as you go” basis


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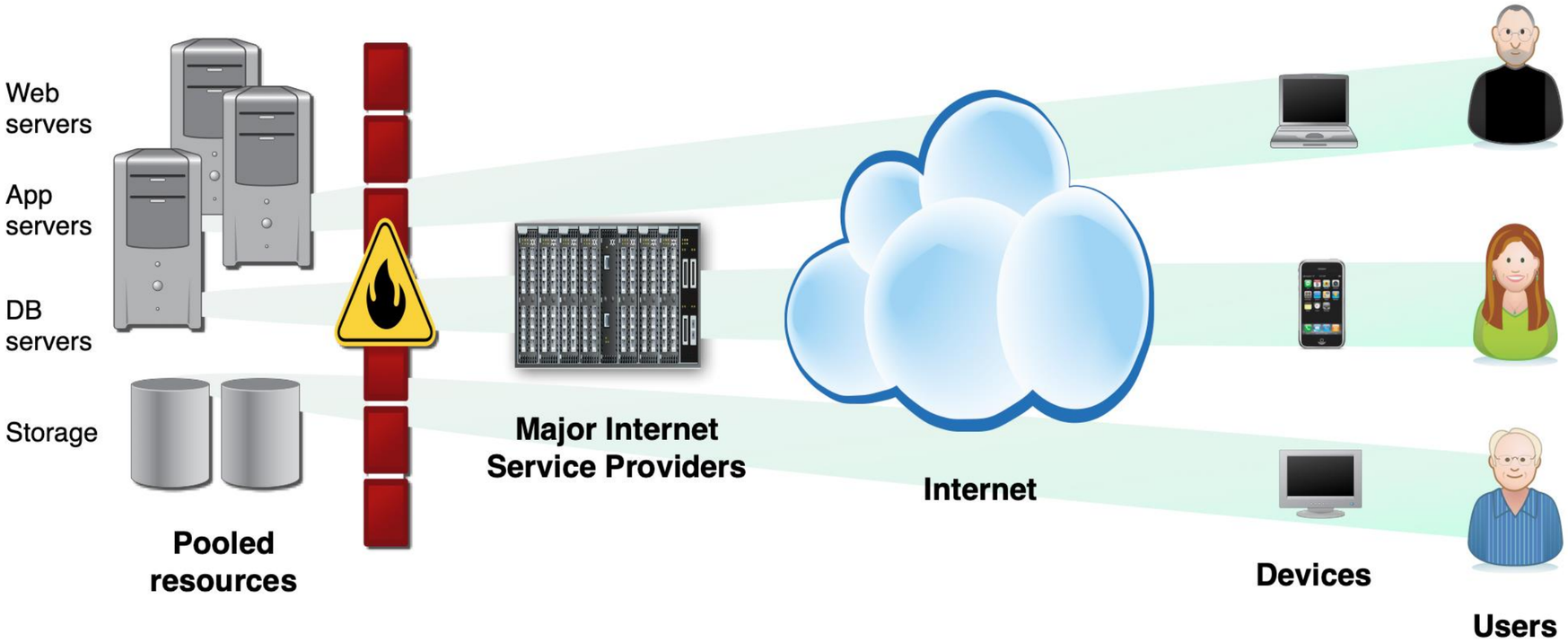

the cloud



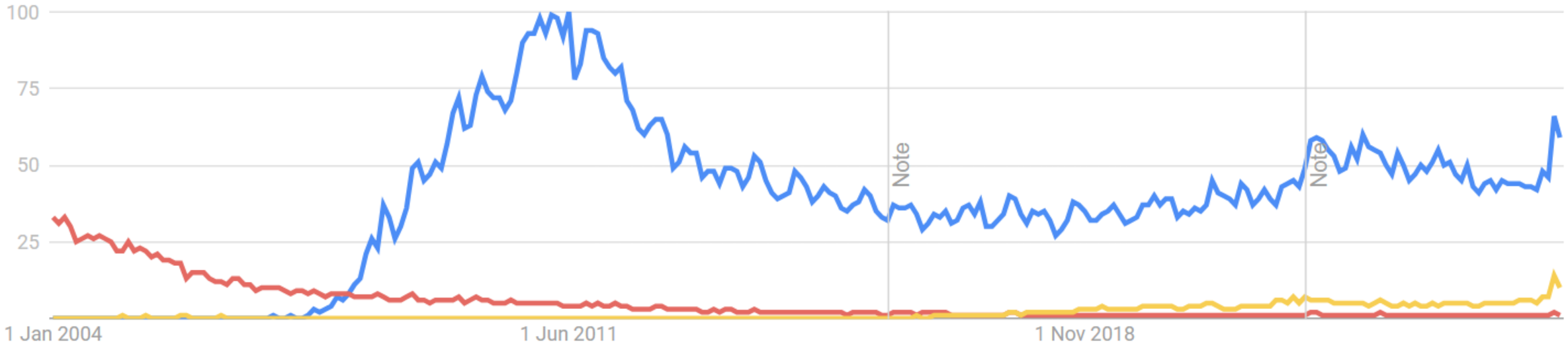
with the Cloud, you can …

- access your documents from any devices (your desktop, laptop and your smartphone)
  - anywhere
  - anytime
- without the hassle
  - to install complex software (well, a simple client program is expected for some services, such as dropbox)
  - to worry about wether the hosting server is running or not

The “cloud”


Web servers
App servers
DB servers
Storage
Pooled resources
Major Internet Service Providers
Internet
Devices
Users

Google Trends 2004 - present



more formal definition of Cloud

- "… a style of computing in which scalable and elastic IT-enabled capabilities are delivered as a service to external customers using Internet technologies."

- "… a standardised IT capability (services, software, or infrastructure) delivered via Internet technologies in a pay-per-use, self-service way."

- "… is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."

The key distinguishing characteristics of cloud

- Resource pooling
- Broad network access
- On-demand self-service
- Measure service
- Rapid elasticity

source: The NIST Definition of Cloud Computing.
U.S. National Institute of Standards and Technology Special Publication 800-145 (September 2011).
http://csrc.nist.gov/publications/nistpubs/800-145/SP800-145.pdf

Resources pooling

- provider's computing resources are pooled to serve multiple consumers using multi-tenant model
- dynamically assigned and reassigned resources according to consumer demand
  - dynamically scalable according to needs
- analogy: bus vs car
  - higher capacity, higher utilisation, lower cost

Broad network access

- off premises: resources are available over the network
- can be accessed by multiple devices, including smartphones, tablets, laptops and desktops
- e.g. Google Drive
  - user can upload a document using the desktop in the office, and access it back at home using iPad.

On-demand self-service

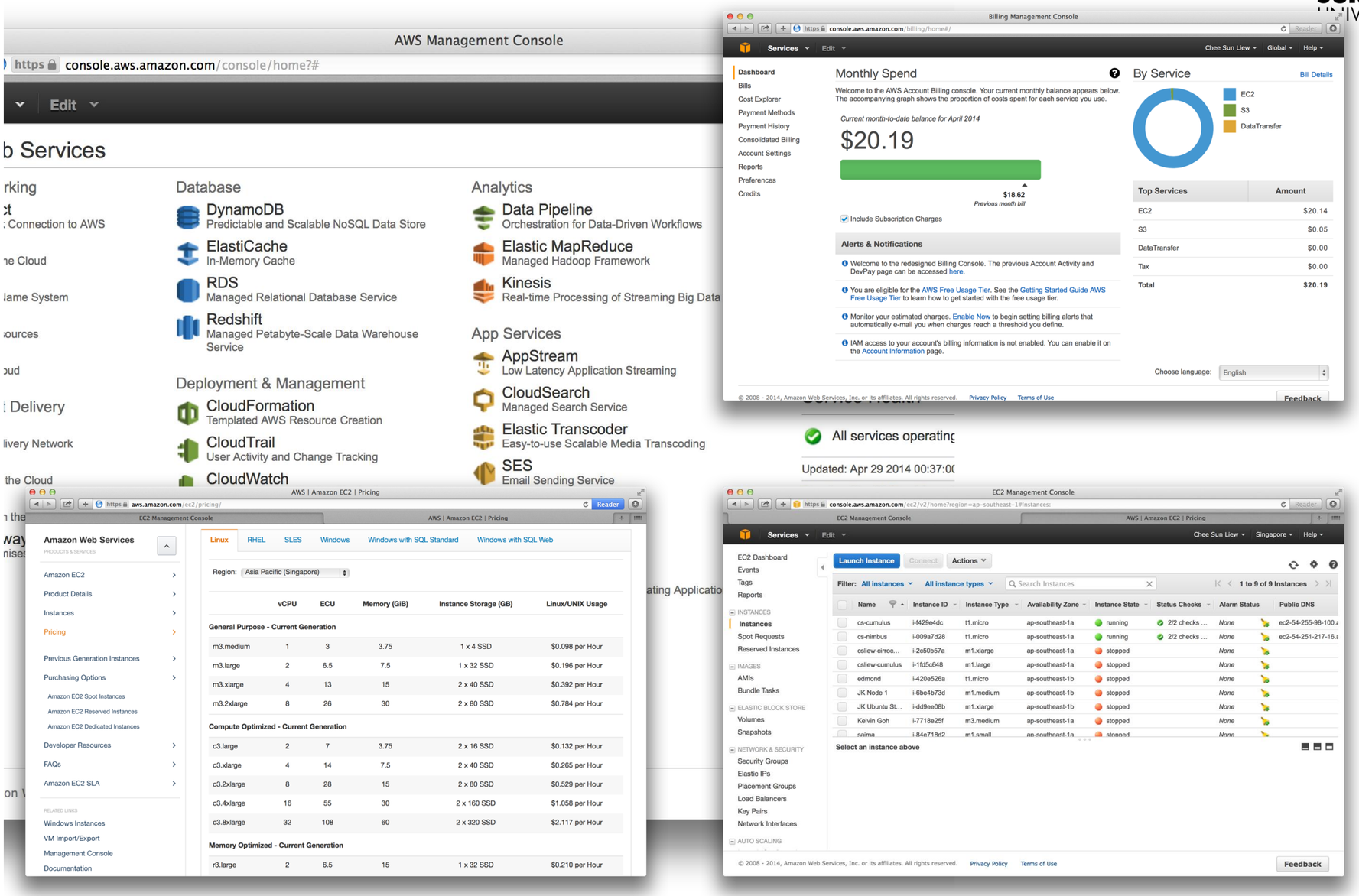
- user can provision the resources, such as server time and storage, as needed without requiring human intervention
- resources can be used as necessary and only when necessary
- e.g. AWS EC2
  - user can initiate any number of virtual servers at anytime of the day, through the user-friendly portal

Measure service (metered billing)

- Cloud systems automatically control and optimise resource use by leveraging a metering capability on a pay-per-use basis.
- Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilised service.
- e.g. in the AWS, user can set usage alarm on the virtual server, which will automatically send out an notification email to the user. User will receive an itemised bill each month.

Rapid elasticity

- Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.
- To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.



source: Snapshot taken from the AWS web service

Who needs cloud?



Your mobile needs it

- state-of-the-art gadget, but still need the cloud:
  - to backup and access your data,
    e.g. DropBox, iCloud
  - to run the heavy backend-processing of most apps,
    e.g. Google map, waze
  - to update your apps
  - and most of your daily usages



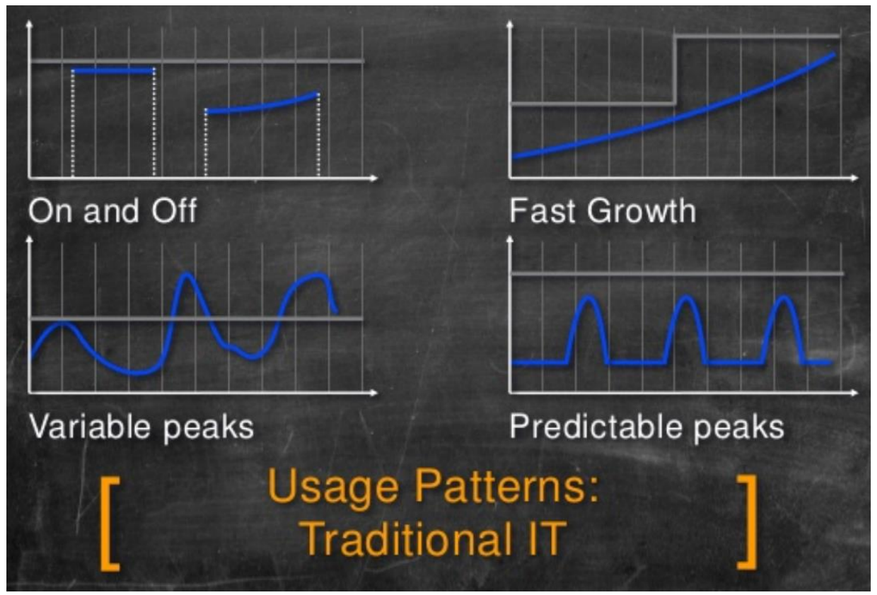
source: www.lenovo.com

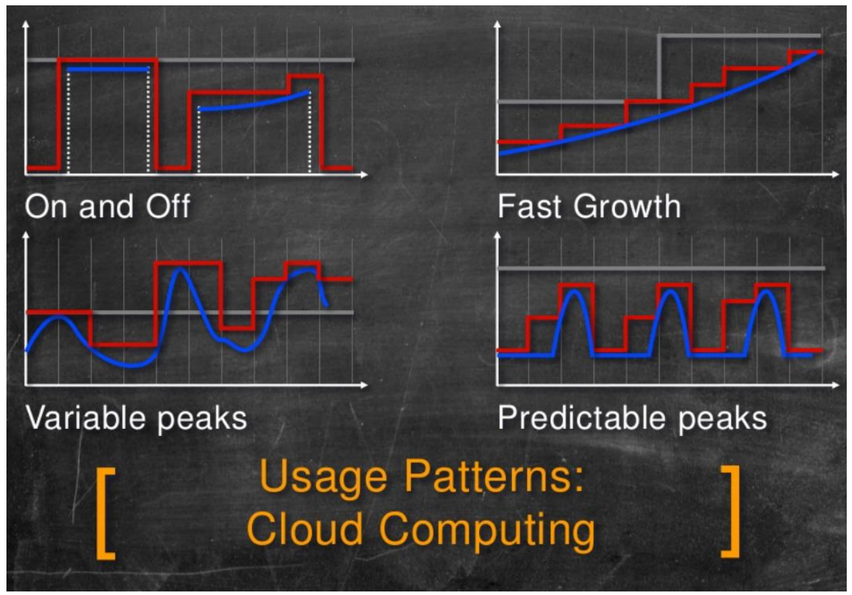
Your company needs it

- lower Total Cost of Ownership (TCO)
  - includes direct and indirect cost of the IT systems:
    - computer hardware and software, e.g. purchasing study, installation and integration, licensing, warranties, migration expenses, and risks
    - operations expenses, e.g. infrastructure (building), electricity (power up and cooling), downtime and failure, backup and recovery, security, and training
    - long-term expenses, e.g. replacement, upgrading, and decommissioning
- measurable benefits:
  - on-demand and pay-as-you-go
  - add/remove IT resources at fine-grained level
  - abstraction of infrastructures

- increase availability and reliability
  - 24/7 availability
  - 99.999999999% durability of Amazon S3
    - corresponds to an average annual expected loss of 0.000000001% of objects.
    - e.g. if you store 10,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000,000 year

- elastic and on-demand
  - easily scalable and pay-as-you-go

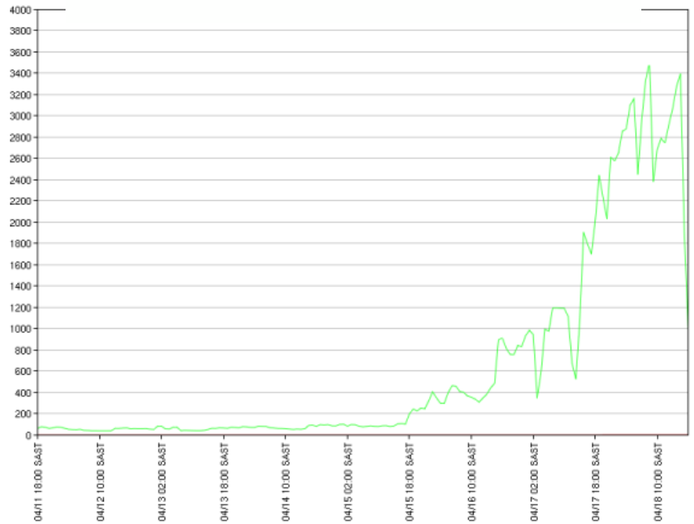




source: AWS 101 - Journey to the AWS Cloud Series @ slideshare

- Animoto (http://animoto.com)
  - user upload a series of photo and the web site turn it into a unique and interesting movie
  - after introduce Facebook apps:
    - 25,000 on Monday, and attracting 750,000 new users in three days
      [signing up 20,000 new users per hour at peak]
    - 50 AWS instances on Monday, and go up to 5000 instances in three days





source: https://aws.amazon.com/solutions/case-studies/animoto/

Cloud computing
is more than
a model for elastic workloads,
cost reduction,
capital preservation and improved
IT efficiency

It’s a revolution!

- It’s a major driver of your business innovation

- expanding business
  - multinational company
  - single infrastructure to be used across the globe

- exploring new business opportunities
  - reaching more users
  - opening up new customer segments and markets
  - diversify business

Malaysia towards a top AI Nation

Malaysia's aspiration of being among the top 20 countries in terms of AI

Embracing AI technology is crucial for societal advancement



Talent

The people to drive the innovation and make the change



Technology

The infrastructure including hardware, software and services as the enabler for AI adoption



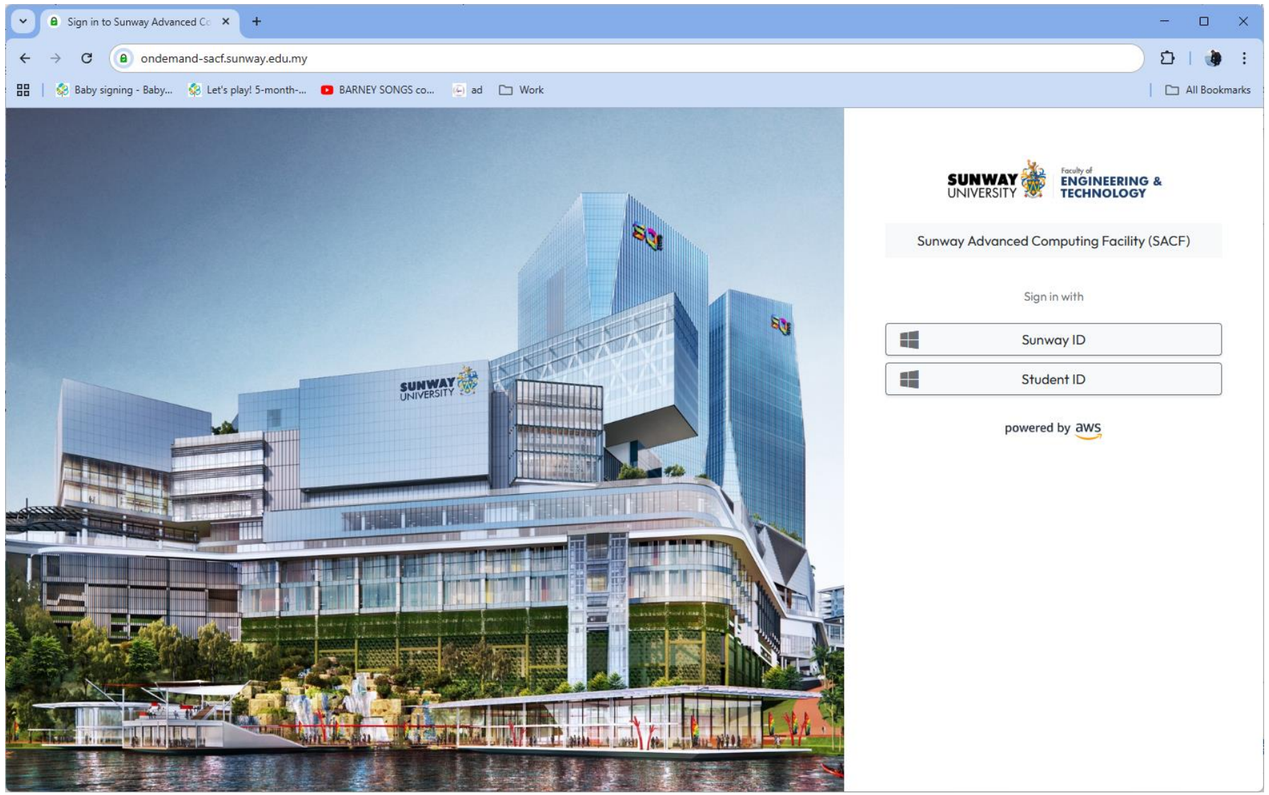
Data

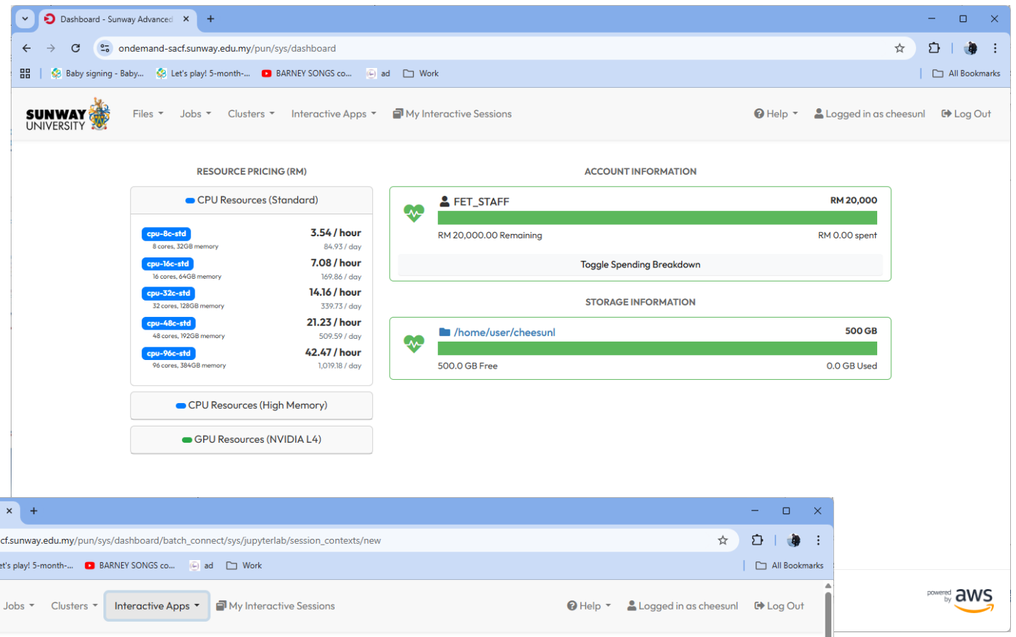
AI needs good data, i.e. quality and quantity

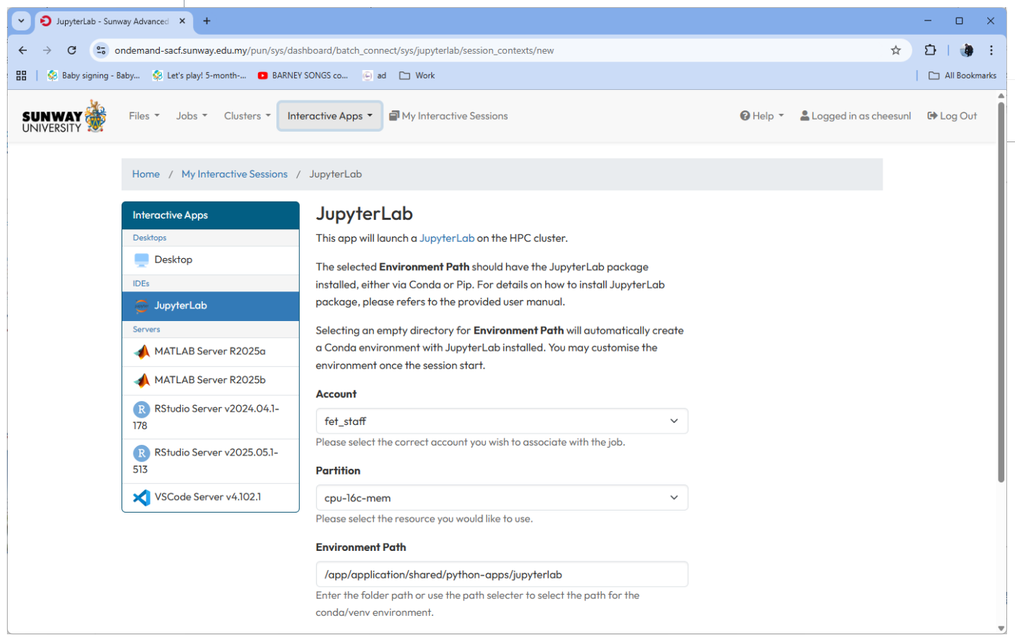
"We cannot, for example, seek to make Malaysia a success as a digital nation while being bound by old and outdated mindsets,"

~ PM Datuk Seri Anwar Ibrahim

Revolutionise Sunway AI research & teaching through AWS – HPC in the Cloud







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People used to believe

1943	"I think there is a world market for maybe five computers." Thomas Watson
1966	“Remote shopping, while entirely feasible, will flop.” Time Magazine
1977	"There is no reason anyone would want a computer in their home." Ken Olsen
1981	“No one will need more than 637KB of memory for a personal computer. 640KB ought to be enough for anybody.” Bill Gates
1995	“I predict the Internet will soon go spectacularly supernova and in 1996 catastrophically collapse.” Robert Metcalfe
2007	“There’s no chance that the iPhone is going to get any significant market share.” Steve Ballmer

Cloud delivery model

- a specific, pre-packaged combination of IT resources offered by a cloud provider.
- three common cloud delivery models have become widely established and formalised:
  - Infrastructure-as-a-Services (IaaS)
  - Platform-as-a-Services (PaaS)
  - Software-as-a-Services (SaaS)

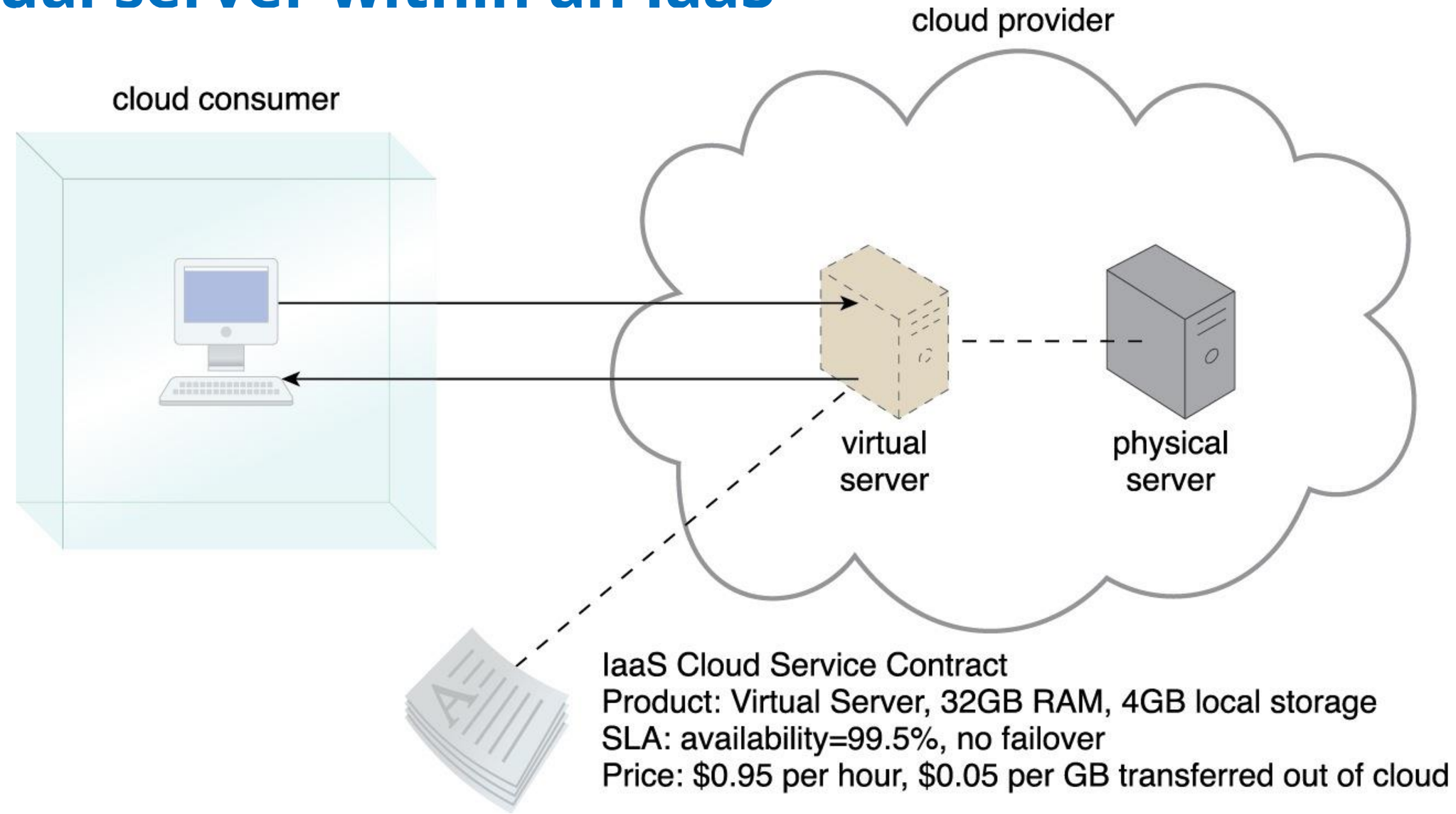
IaaS

- a self-contained IT environment comprises infrastructure-centric IT resources
  - hardware, connectivity and operating systems
- can be accessed and managed via cloud service-based interfaces and tools
- IT resources are typically virtualised and packaged into bundles that simplify up-front runtime scaling and customisation of the infrastructure.

Characteristic of IaaS

- the IT resources are generally not pre-configured
- this placing the administrative responsibility directly upon cloud consumer
- used by cloud consumers that require a high level of control over their cloud-based infrastructure

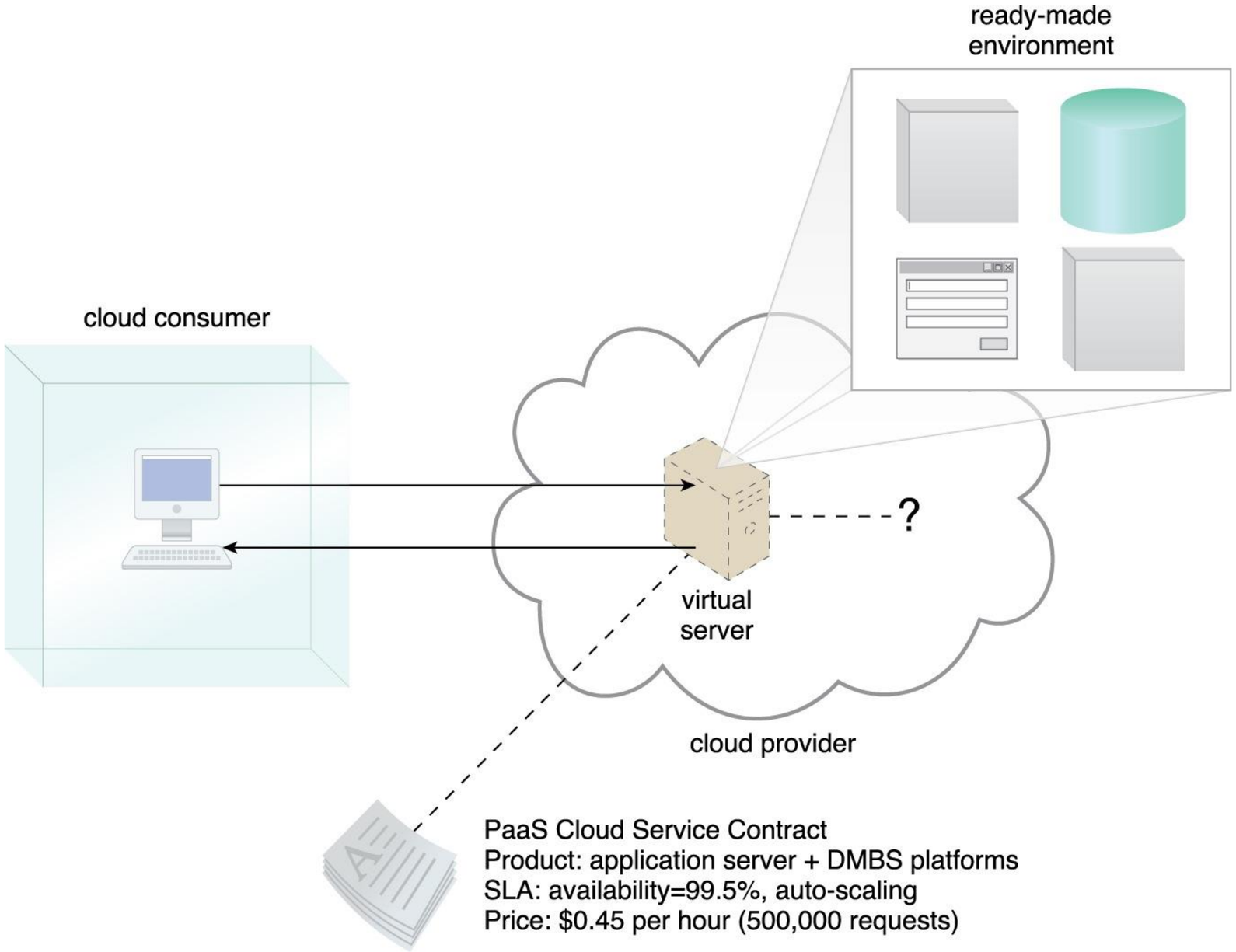
Virtual server within an IaaS



PaaS

- a predefined ready-to-use environment comprises already deployed and configured IT resources.
- relied on the usage of a ready-made environment
  - establishes a set of pre-packaged products and tools to support the delivery life-cycle of custom applications

Ready-made PaaS environment



Why a cloud consumer needs PaaS?

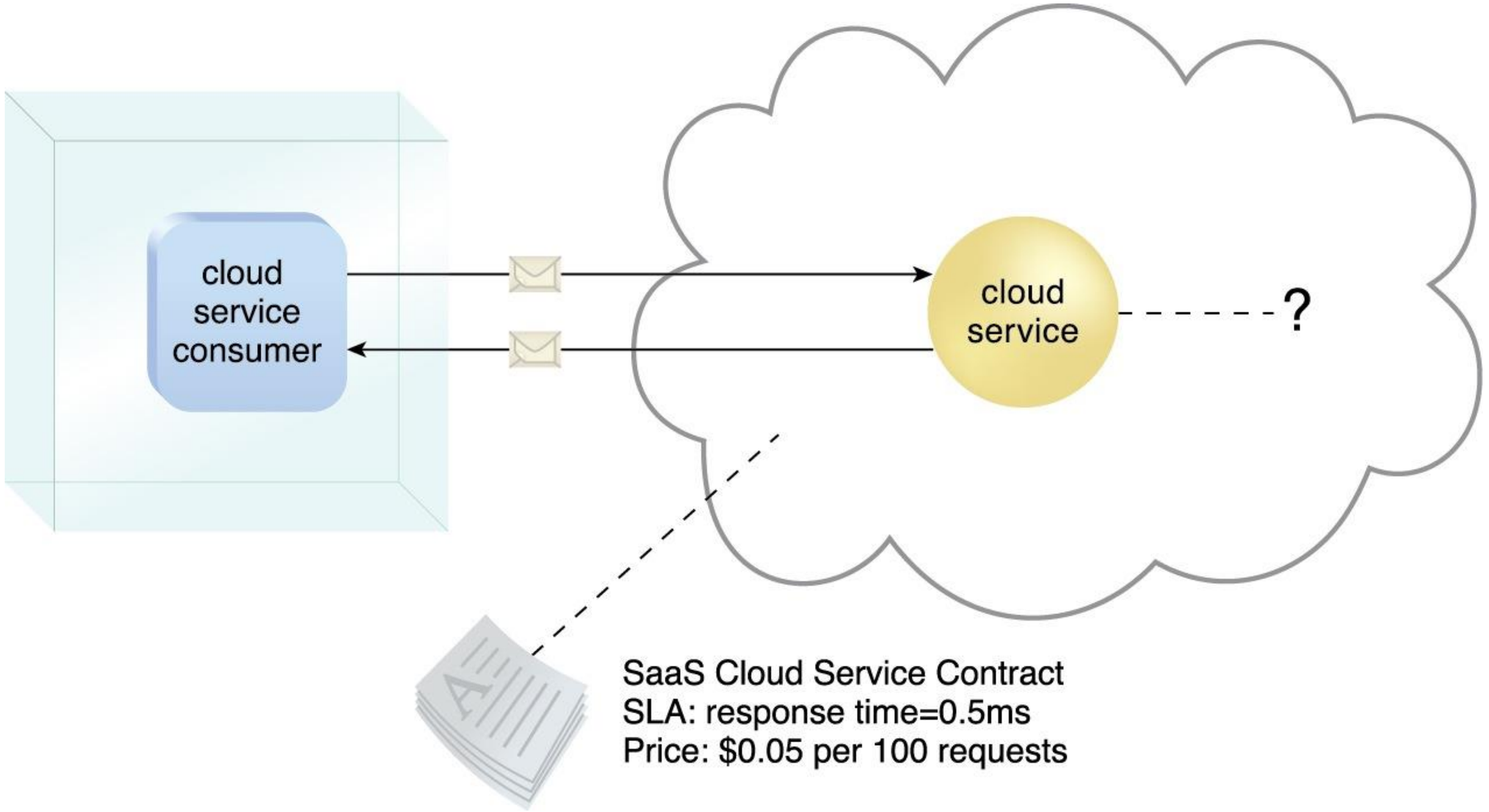
- To extend on-premise environments into the cloud
  - scalability and economic purpose
- To substitute an on-premise environment
- To become a cloud provider and deploys its own cloud services to other external cloud consumers

- *the cloud consumer is spared the administrative burden of setting up and maintaining bare IT infrastructure*

SaaS

- a software program positioned as a shared cloud service and made available as a product or generic utility
- used to make a reusable cloud service widely available to a range of cloud consumer
- the entire marketplace exists around SaaS products can be leased and used for different purposes via different terms

Cloud service in SaaS



Characteristics of SaaS

- cloud consumers are granted limited administrative control over SaaS implementation
- a PaaS cloud consumer can use the development environment to build a SaaS and made available to other cloud consumers
  - a PaaS cloud consumer become cloud providers

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A comparison of control levels

Cloud Delivery Model	Typical Level of Control Granted to Cloud Consumer	Typical Functionality Made Available to Cloud Consumer
SaaS	usage and usage-related configuration	access to front-end user interface
PaaS	limited administrative	moderate level of administrative control over IT resources relevant to cloud consumer's usage of platform
IaaS	full administrative	full access to virtualized infrastructure-related IT resources and, possibly, to underlying physical IT resources

A comparison of activities

Cloud Delivery Model	Common Cloud Consumer Activities	Common Cloud Provider Activities
SaaS	uses and configures cloud service	1. implements, manages, and maintains cloud service 2. monitors usage by cloud consumers
PaaS	develops, tests, deploys, and manages cloud services and cloud-based solutions	1. pre-configures platform and provisions underlying infrastructure, middleware, and other needed IT resources, as necessary 2. monitors usage by cloud consumers
IaaS	sets up and configures bare infrastructure, installs, manages, and monitors any needed software	1. provisions and manages the physical processing, storage, networking, and hosting required 2. monitors usage by cloud consumers

Cloud Deployment Model

- represents a specific type of cloud environment, primarily distinguished by ownership, size, and access.
- there are four common cloud deployment models:
  - public clouds
  - community clouds
  - private clouds
  - hybrid clouds

Public clouds

- publicly accessible cloud environment owned by third party cloud provider
- generally offered to cloud consumer at a cost
- cloud provider is responsible for the creation and on-going maintenance of the cloud and its IT infrastructure
- e.g. AWS and Salesforce

Characteristic of public cloud

- OSSM (awesome) by Dave Nielsen
  - **O**n-demand
    - the infrastructure or hardware is already setup and ready to be deployed
  - **S**elf-service
    - the customer chooses what they want, when they want it
  - **S**calable
    - customers can choose how much they want and ramp up if necessary
  - **M**easurable
    - there's metering/reporting so you know you are getting what you pay for

Public cloud benefits

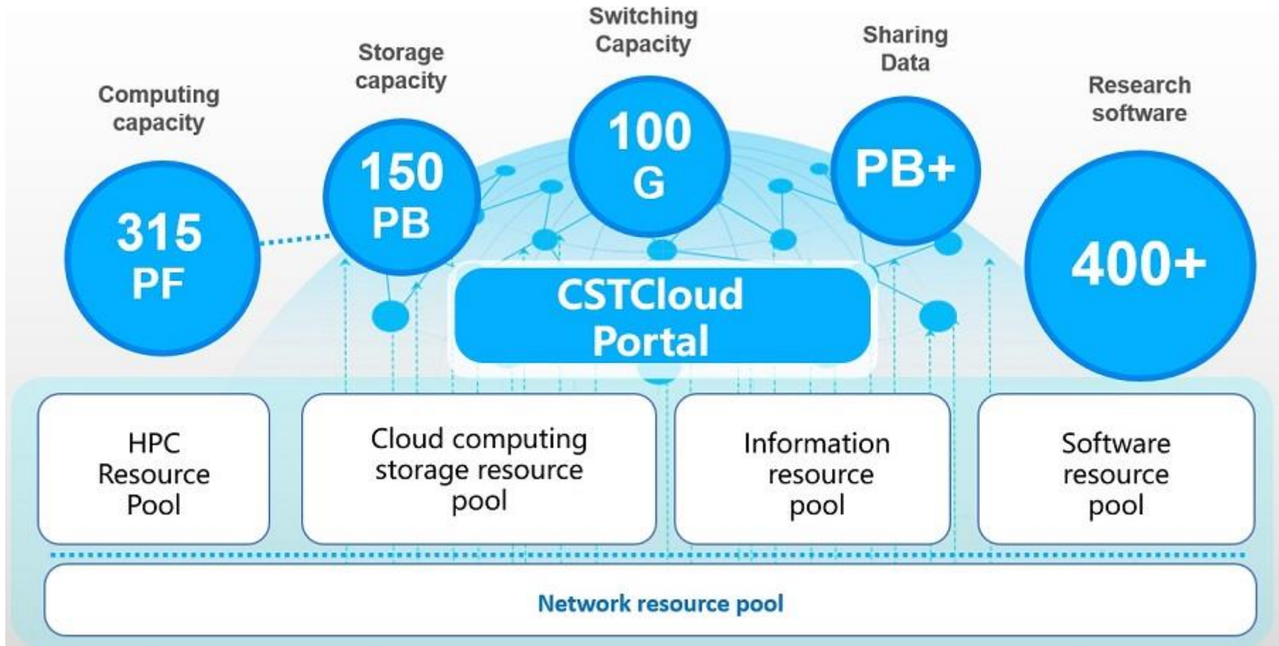
- Cost Savings
  - less expensive (TCO perspective) than traditional on-premise computing and will often be cheaper than Private Cloud or Hybrid Cloud
- Low, or no, Upfront Costs
- Reduced Infrastructure Footprint
  - no physical infrastructure footprint for Public Cloud customers
- Increased Flexibility
- Software that can be Decoupled from Infrastructure Constraints
  - by moving to the Public Cloud, organisations can scale software without concerns about related infrastructure needs
- Capacity Planning
  - virtually unlimited capacity as and when an organisation needs it

Case study: InvestLab on AWS

- InvestLab is a financial service technology company
- develop trading, market data and social research products that enable individual investors and brokers to access global market
- challenge: fast, on-demand analysis of time-sensitive and volatile market data
  - < 5 minutes in response to opening market and global event
  - 80% - 90% idle computing resources
- benefit: 40% reduction in the fixed cost of launching a software product

Community clouds

- similar to public clouds, but only limited to a specific community of cloud consumer
- may be owned by the community or third party cloud provider
- members do not necessary guarantee access/control the IT infrastructure
- e.g. Chinese Academy of Sciences, China Science and Technology Cloud (https://www.cstcloud.net/)



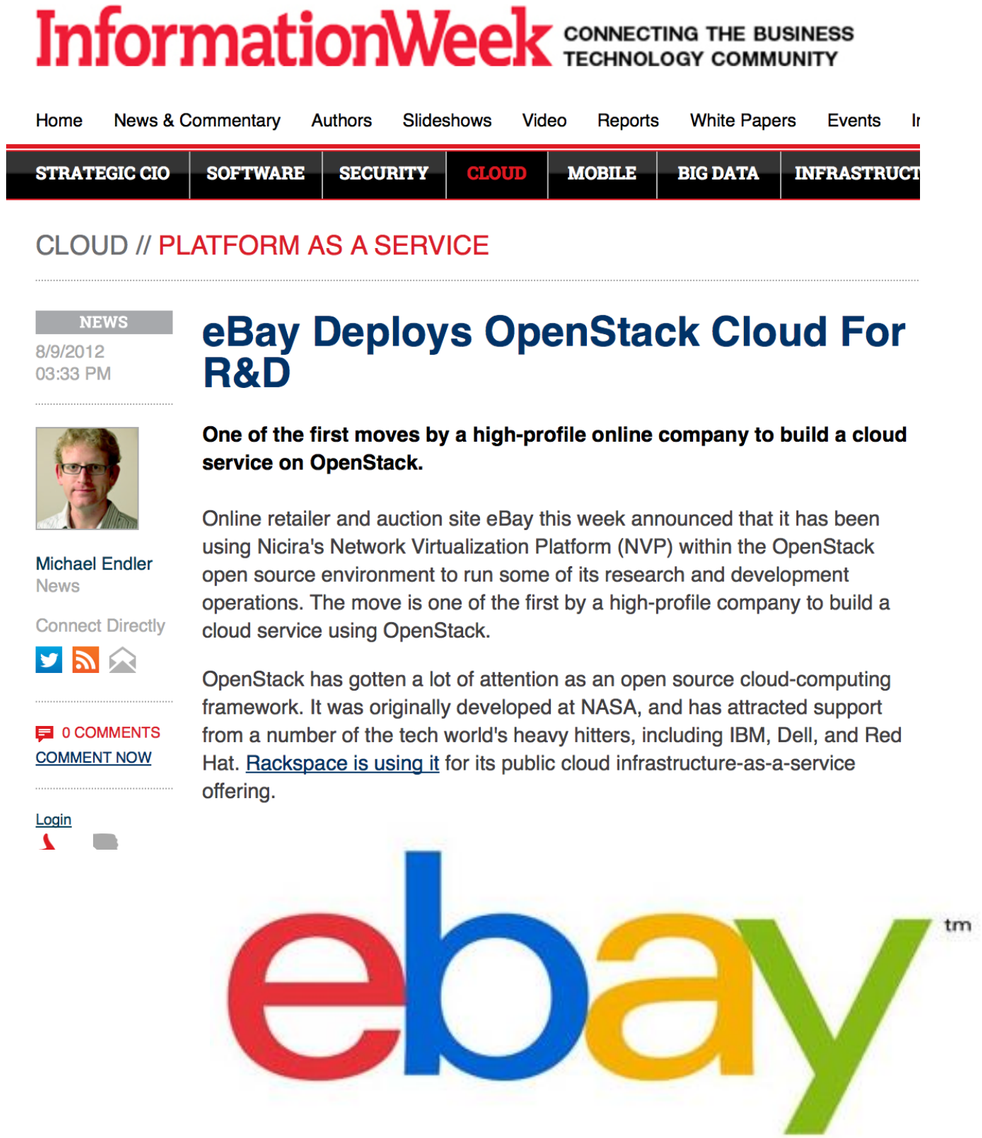
中国科技云
China Science & Technology Cloud

Private clouds

- owned by a single organisation
- use cloud computing technology to provide centralised access to IT resources by all parts
  of the organisation
- the same organisation is technically both the cloud consumer and cloud provider
- e.g. eBay Cloud

The five key steps in creating private cloud

- simplify
  - process, technology and organisational
- automate everything
- applications anywhere
- plan for failure
  - Murphy's law "Anything that can go wrong will go wrong"
- proportional security
  - finer control of security



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Michael Endler
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eBay Deploys OpenStack Cloud For R&D

One of the first moves by a high-profile online company to build a cloud service on OpenStack.

Online retailer and auction site eBay this week announced that it has been using Nicira's Network Virtualization Platform (NVP) within the OpenStack open source environment to run some of its research and development operations. The move is one of the first by a high-profile company to build a cloud service using OpenStack.

OpenStack has gotten a lot of attention as an open source cloud-computing framework. It was originally developed at NASA, and has attracted support from a number of the tech world's heavy hitters, including IBM, Dell, and Red Hat. Rackspace is using it for its public cloud infrastructure-as-a-service offering.

ebay tm

Four approaches to private cloud



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Private cloud benefit

- In addition to the generic cloud benefits, private cloud offers additional benefits:
  - organisations whose legislative or regulatory requirements demand a physical or logical separation from the data of other organisations
  - organisations with existing infrastructure that they wish to continue using, but with increased efficiencies
  - organisations that have specific requirements in terms of a technology stack
  - Can reduce the risks associated with vendor lock in

Concerns about deploying private cloud

- private clouds are small scale
  - volume drives cost
- legacy applications don’t easily cloud-ready
  - marginal improvements at best
- on-premise does not mean more secure
  - unless your company plan to spend more money than Amazon and Google on cloud security
- no trick than can be borrowed from the giant
  - Amazon or Google spent years on optimising the operation of their public cloud, and your innovation rate never the same

Hybrid clouds

- a cloud environment comprised of two or more different cloud deployment models
- deploy cloud services processing sensitive data to a private cloud and other, less sensitive cloud services to a public cloud
- e.g. Action for Children on Rackspace

Hybrid cloud benefits

- Flexibility
  - retain specific parts of organisation IT infrastructure while moving some aspects (archiving or disaster recover) to the cloud
- Maximising ROI on Existing Infrastructure
  - organisations to continue to utilise their existing infrastructure while moving excess workloads or specific parts of their operation to the cloud.
- Compliance
  - organisations in particular industries can retain particular workloads on-premise or on dedicated infrastructure.

When to use hybrid cloud

- Where the organisation has legacy applications that do not make sense to move to the public cloud
- Where compliance requirements create a demarcation between what can be stored on the public cloud and what must remain locally
- Where peaks in traffic (e.g. after specific marketing campaigns) demand that traffic be “burst” to the public cloud
- Where the cost of having backups on-premise for disaster recovery is prohibitive
- Where organisations wish to prototype applications cheaply before deploying them on in-house infrastructure
- Where avoiding vendor lock-in is a primary concern
- Where web applications require high database performance but need to

Public Cloud



Shared • Open • Pay-per-use

An easily deployed virtual server on a shared hosting infrastructure.

Quickly or automatically scale your servers according to workload.

Pay-per-use for unlimited computing resource, storage and bandwidth.

➔ Choose a cloud with quality service, engineering and infrastructure!

➔ Choose carefully to avoid “noisy neighbor” effect from sharing!

Private Cloud



Dedicated • Secure • Fixed Cost

A secure virtual data center on your own dedicated infrastructure.

Servers and storage can be added and scaled according to your needs.

Planned resource capacity with fixed costs and unmetered bandwidth.

➔ Financial control and quick IT response to business needs.

➔ Having IT control means you also need IT expertise and resources!

Hybrid Cloud



Pick from the best of both worlds

A mix of dedicated infrastructure and public cloud infrastructure.

Cherry-pick the best infrastructure for your different services.

Keep data secure and control costs and resources with dedicated infrastructure.

➔ Choose public or private cloud for each service according to its performance profile!

➔ Requires IT architecture planning, support and expertise!

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Summary

- What did I learn in this module?
  - The concept of computing as a service
  - The definition of cloud computing
  - The key distinguishable characteristics of cloud
  - The benefits given by cloud
  - Cloud Delivery Model
  - Cloud Deployment Model

Acknowledgement

The figures and contents used in this lecture are adopted and adapted from:

- van Steen, M. and Tanenbaum A.S. (2023). Distributed Systems, 4th Edition. distributed-systems.net.
- Colouries, G., Dollimore, J., Kindberg, T and Blair, G. (2021). Distributed Systems: Concepts and Design, 5th Edition. Pearson.
- Thomas Erl , Zaigham Mahmood, and Ricardo Puttini (2013). Cloud Computing: Concepts, Technology & Architecture. Prentice Hall.
- Web references.

THANK YOU

Chee Sun LIEW
cheesunl@sunway.edu.my

